

KOBELYAKY, Ukraine

WMO: 336210

Lat: 49.1539N

Lon: 34.2078E

Elev: 118

StdP: 99.92

Time Zone: 2.00 (EUE)

Period: 84-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-16.4	-22.1	0.5	-18.6	-19.5	0.7	-16.2	8.3	-8.9	7.4	-3.3	2.2	180	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	31.7	20.1	30.0	19.2	28.3	18.6	21.6	28.4	20.6	27.3	19.7	26.1	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.3	24.9	18.2	13.3	23.6	17.3	12.5	22.7	63.3	28.3	59.7	27.1	56.6	26.1	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.9	5.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-3.9	-3.2	1.5	10.3	16.9	19.1	21.0	20.9	15.9	8.4	1.6	-2.4		
	DBStd	10.44	5.64	6.18	5.46	4.02	3.94	3.11	3.06	3.39	4.31	4.69	5.65	4.88		
	HDD10.0	1838	431	368	264	46	2	0	0	0	4	86	253	384		
	HDD18.3	3727	689	602	521	240	77	27	10	13	97	307	502	642		
	CDD10.0	1444	0	0	2	56	216	272	342	337	182	38	1	0		
	CDD18.3	292	0	0	0	1	33	49	93	91	24	0	0	0		
	CDH23.3	2622	0	0	0	8	274	379	830	900	229	2	0	0		
	CDH26.7	799	0	0	0	0	44	93	259	339	63	0	0	0		
(14) Wind	WSAvg	2.6	2.8	2.7	3.0	2.8	2.4	2.2	2.3	2.1	2.4	2.4	2.6	2.9		
(15) Precipitation	PrecAvg	521	45	38	32	37	46	55	53	46	35	40	43	50		
	PrecMax	837	127	81	81	88	119	144	123	167	77	186	128	134		
	PrecMin	278	8	2	2	2	13	4	8	2	3	0	4	4		
	PrecStd	127	29	21	19	22	26	31	31	34	23	43	30	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	10.0	17.2	24.3	29.4	31.7	33.0	34.7	31.0	23.9	15.3	9.4		
		MCWB	5.0	8.0	9.7	14.3	17.1	20.5	21.7	19.5	20.1	15.1	12.3	8.1		
	2%	DB	4.2	7.0	13.6	21.4	27.8	29.1	31.2	32.4	28.4	20.5	12.2	6.2		
		MCWB	3.1	5.6	8.3	12.8	16.7	19.7	20.5	19.8	17.9	13.9	10.3	5.0		
	5%	DB	3.0	4.9	10.8	19.4	26.5	27.1	29.4	30.2	25.8	17.9	10.2	4.9		
		MCWB	2.0	3.4	7.1	11.9	16.8	19.0	19.6	19.1	16.5	13.2	8.4	4.0		
	10%	DB	2.1	3.3	8.4	17.4	24.6	25.3	27.6	28.2	23.5	15.4	8.3	2.9		
		MCWB	1.3	2.1	5.4	10.7	16.6	18.2	18.9	18.6	15.7	12.3	6.9	2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	8.1	10.6	15.6	19.9	22.3	23.4	22.5	20.8	17.1	12.9	8.2		
		MCDB	5.6	9.6	15.9	22.8	25.6	28.0	30.8	29.9	29.8	20.8	15.0	9.1		
	2%	WB	3.3	5.9	8.8	13.6	18.7	21.0	21.9	20.9	18.7	15.4	10.5	5.2		
		MCDB	3.9	6.8	13.0	19.6	24.7	27.3	29.2	28.8	25.9	18.2	11.9	6.0		
	5%	WB	2.2	3.6	7.3	12.6	17.9	19.9	20.7	19.9	17.4	14.0	8.7	3.8		
		MCDB	2.6	4.5	10.3	18.1	23.9	25.7	26.7	27.9	24.3	16.6	9.8	4.9		
	10%	WB	1.4	2.2	5.7	11.4	17.0	18.9	19.7	19.0	16.2	12.4	7.0	2.2		
		MCDB	1.9	3.1	8.0	16.4	22.7	23.9	25.5	26.3	22.1	15.4	8.3	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.5	6.9	9.1	10.2	9.6	10.4	11.0	10.0	8.2	5.4	4.2		
		MCDBR	4.4	6.0	11.1	11.8	12.4	11.7	12.7	13.8	13.5	11.7	6.8	5.4		
	5% WB	MCWBR	4.1	5.0	6.8	5.9	5.0	5.3	4.7	5.1	5.7	6.7	5.1	4.8		
		MCWBR	4.0	5.5	10.1	10.5	10.7	10.1	10.8	12.0	12.6	9.3	5.8	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.353	0.384	0.448	0.414	0.406	0.437	0.432	0.405	0.367	0.351	0.326		
		taud	2.195	2.151	2.196	2.084	2.267	2.310	2.228	2.228	2.272	2.353	2.301	2.271		
	Ebn at Noon		700	769	815	797	847	856	820	804	786	754	668	647		
		Edh at Noon	79	105	119	148	129	124	133	127	110	86	72	66		
(44) All-Sky Solar Radiation	RadAvg		0.94	1.78	2.98	4.43	5.79	6.21	6.09	5.44	3.72	2.18	0.93	0.63		
	RadStd		0.18	0.32	0.42	0.41	0.49	0.54	0.37	0.37	0.49	0.32	0.17	0.14		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)		+0.76	N/A	N/A	+1.04	N/A	N/A	-147	-200	+122	+77	

Nomenclature: See separate page